q,t-Catalan and hyperplane arrangement

This seminar includes four topics

(i) q,t-Catalan vs hyperplane arrangement

(ii) Hilbert series for diagonal harmonic vs hyperplane arrangement

(iii) Preliminaries for the proof, the proof

(iv) Type B_n q,t-Catalan, Type B_n Shi arrangement, conjectures and open problem.

Main references:

- [Armstrong, 2013, JoC]
- [Haglund, 2004, Lecture notes]
- [Lingxi Lu, 2019, Master thesis]
- [A. Oblomkov, Oct. 2026, arxiv]

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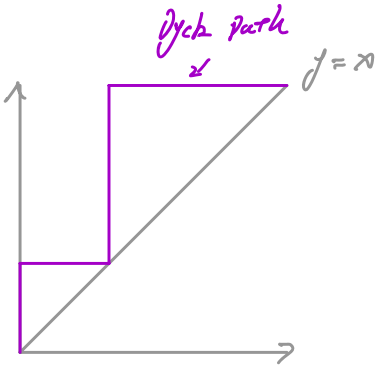
First class: Introduction to classical Catalan

Catalan number:

- $C_n = \frac{1}{n+1}\binom{2n}{n}$ denote the n-th Catalan number.
- The recursion is: $C_n = \sum_{k=1}^{\frac{n}{2}} C_{k-1} C_{n-k}$.
- [sta 99] for a list of over 200 objects counted by the Catalan numbers.

Here we introduce two objects: "Dyck path" and "hyperplane arr".

(i) $L^{+}_{n,n}$: the set of lattice path from $(0,0) \to (n,n)$, which only takes steps $N = (0,1) \uparrow$, $E = (1,0) \to$, and never go below the line $y = x$. Elements in $L^{+}_{n,n}$ are called "Dyck paths"



Then, $\#L^{+}_{n,n} = C_n$.

proof: Let $L_n = \#L^{+}_{n,n}$. Then the number of Dyck path from $(0,0) \to (k,k)$ which never touch $y = x$ except $(0,0)$ and (k,k) is L_{k-1}. Hence:

$$L_n = \sum_{k=1}^{n} L_{k-1} \cdot L_{n-k}$$

(ii) Let $V = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 0\}$ and

$$Shi(n) = \{x_i - x_j = a : 1 \le i < j \le n,\ a = 0, 1\}.$$

be the Shi arrangement. The connected components of $V - \bigcup_{H \in Shi(n)} H$ are called chambers. Let

$$C_0 = \{x \in V : x_1 > x_2 > \cdots > x_n\}$$

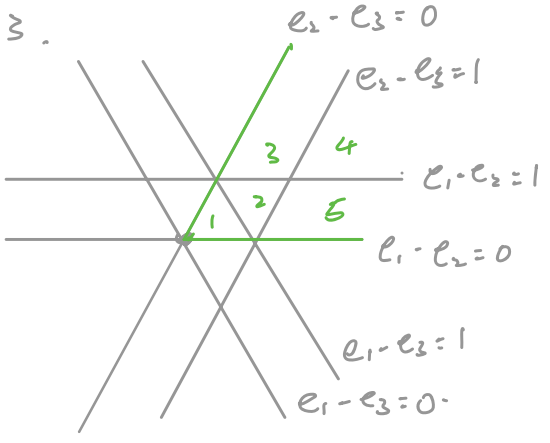
be the dominant cone. Then:

"# chambers of $Shi(n)$ that lie in C_0"

$$= \frac{1}{n+1}\binom{2n}{n}$$

$$= C_n \text{ the classical Catalan number}$$

for example: for $n = 3$.



$$5 = \frac{1}{3+1}\binom{6}{3}$$

The proof: "find a bijection between chambers and Dyck paths"

Let $\Phi^+ = \{e_i - e_j : 1 \le i < j \le n\}$ be positive roots
$\Delta = \{e_i - e_{i+1} : 1 \le i < n\}$ be simple roots
} an type A_{n-1} root system

For $\alpha, \beta \in \Phi^+$, $\alpha \le \beta \iff \beta - \alpha \in \mathbb{N}\Delta$.

$I \subseteq \Phi^+$ is called an ideal, if "$\alpha \in I$, $\alpha \le \beta$" $\Rightarrow$ "$\beta \in I$".

shi chambers those in the dominant cone

$\updownarrow$ ①

root ideal $I \subseteq \Phi^+ = \{e_i - e_j : 1 \le i < j \le n\}$.

$\updownarrow$ ②

Dyck paths in $L^+_{n,n}$.

pf ①: Every shi chamber C in the dominant cone, define

$$I_C = \{e_i - e_j : 1 \le i < j \le n,\ \forall x \in C,\ x_i - x_j > 1\}.$$

This gives bijection ①.

pf ②: we draw Φ^+ in the following way: "square ij represent" root $e_i - e_j$.

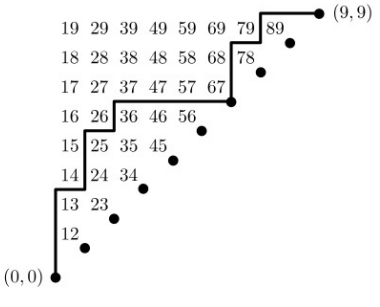


Figure 3.1: An ideal and its corresponding Dyck path.

Then $\alpha \le \beta \Leftrightarrow \beta$ occurs weakly to the left and weakly above α.

Hence it's easy to see the bijection ②.

First class : q,t - Catalan number

There are two q-analogue of Catalan number base on statistics on Dyck paths.

(i) MacMahon introduced a $\mathrm{maj}(\sigma(w))$ statistic of $w \in L_{n,n}^{+}$, (1915 ~). Then,

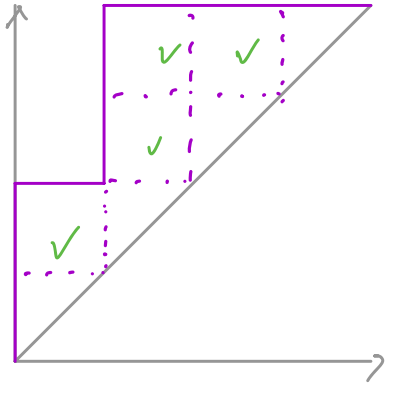
$$\sum_{w \in L_{n,n}^{+}} q^{\mathrm{maj}(\sigma(w))} = \frac{1}{[n+1]_q} \begin{bmatrix} 2n \\ n \end{bmatrix}_q .$$

where

$$[k]_q = \frac{1-q^k}{1-q}, \quad [k]_q! = [1]_q \cdots [k]_q, \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]_q!}{[k]_q! \, [n-k]_q!}$$

(ii) [Carlitz, Riordan, 1964, Duke Math. J].

Define "$\mathrm{area}(w)$" of $w \in L_{n,n}^{+}$ to be the number of squares that entirely lie between w and $y=0$.



$\mathrm{area}(w) = 4$.

They define

$$C_n(q) = \sum_{w \in L_{n,n}^{+}} q^{\mathrm{area}(w)}$$

Prop: $C_n(q)$ has recursion:

$$C_n(q) = \sum_{k=1}^{n} q^{k-1} C_k(q) C_{n-k}(q) .$$

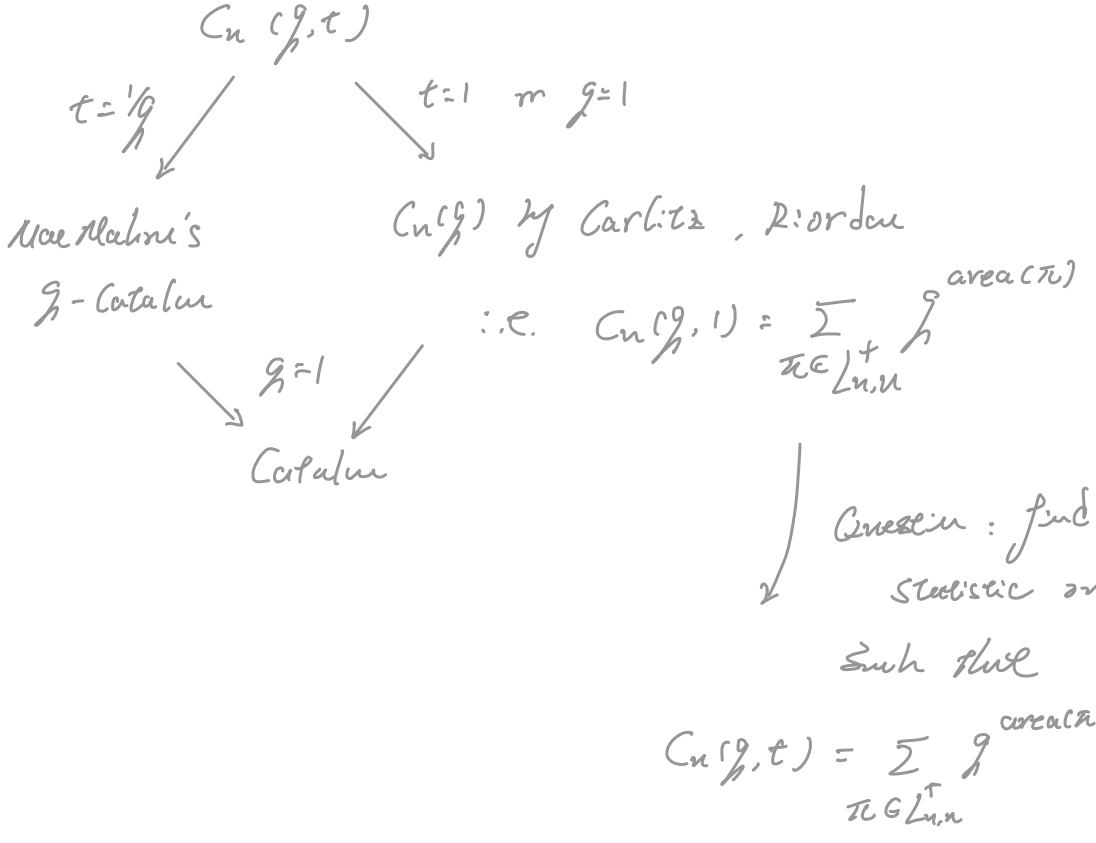
q,t - Catalan:

[Garsia, Haiman, 1996, J. Algebraic Combin.]

They defined

$$C_n(q,t) = \mathcal{H}(\mathcal{D}\mathcal{H}_n^{\varepsilon}; q, t),$$

and showed that



☆ Combinatorial formula for $C_n(q,t)$.

Two statistic on Dyck path: area, bounce

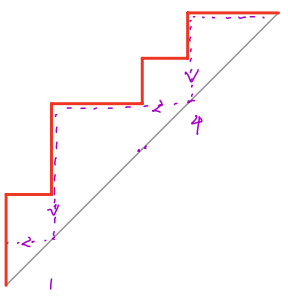
• Def: "bounce statistic" [Haglund, 2003, Adv. math].

Launch a bouncy ball from (n,n) to $(0,0)$

it will strike the diagonal at places

(n,n), (j_1, j_1), ..., (j_b, j_b)

$(0,0)$.



$$\text{bounce}(\pi) := \sum_{i=1}^{b-1} j_i$$

• (rejected by Haglund in 2000 [Adv], proved by Garsia and Haglund few years later by using plethysm of Macdonald polynomials. [Discrete math]; PNAS

$$H(DH_n^{\varepsilon}; q, t) = C_n(q,t) = \sum_{\pi \in L_{n,n}^{+}} q^{\text{area}(\pi)} t^{\text{bounce}(\pi)}.$$

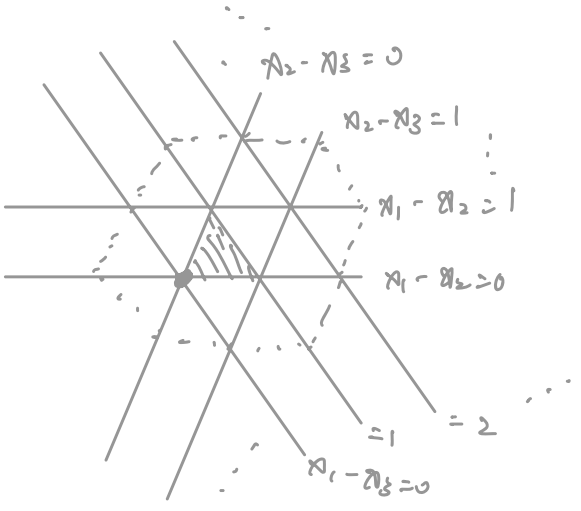
2-wise class: statistic on hyperplane arrangement and q,t-Catalan in terms of hyperplane arrangement.

Basic concepts:

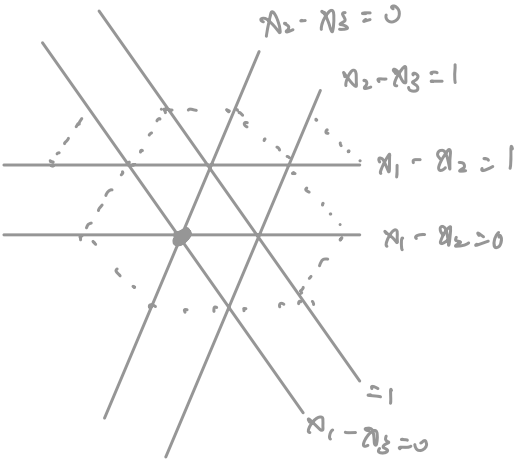
• chambers (i.e. connected component of the complement of the hyperplane arrangement) of

$$\text{Aff}(n) = \{ x_i - x_j = a : 1 \le i < j \le n, a \in \mathbb{Z} \}$$

are called alcoves.



$\mathcal{A}ff(3)$



$Shi(3)$

Let $A_0 = \{ x : x_1 > x_2 > \cdots > x_n > x_1 - 1 \}$ be the fundamental alcove.

Each chamber of $Shi(n)$ contains a set of alcoves.

- "Shi alcove". For each chamber of $Shi(n)$, there is a unique alcove of minimum length. We call this alcove the Shi alcove.

- Shi alcoves in dominant cone are called positive Shi alcoves.

- We now have bijections:
  positive Shi alcoves $\mathcal{A}$ $\longleftrightarrow$ positive chambers C
  $\longleftrightarrow$ root ideal $I \subseteq \Phi^+$
  $\longleftrightarrow$ Dyck paths π

The two statistics : For each **positive** chi arrow A, define statistics

- $shi(A) :=$ # shi hyperplanes separating A from A_0.

Then, ✡ $area(n) = \binom{n}{2} - shi(A)$

- $ish(A) :=$ # ish hyperplanes separating A from A_0

where

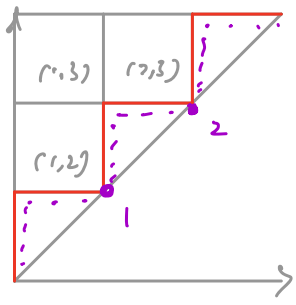
$$ish(n) = cone(n) \cup \{ x_i - x_n = a : 1 \le i < n, a = 1, \dots, n-i \}$$

Then, ✡ $bounce(n) = ish(A)$.

Consequently, by Haglund's combinatorial formula,

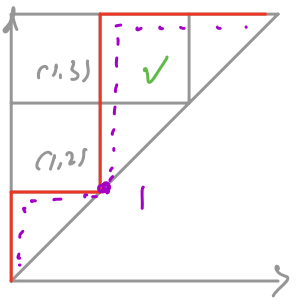
$$C_n(q,t) = \sum_{A : \text{positive shi arrow}} q^{\binom{n}{2} - shi(A)} t^{ish(A)}$$

Let's prove in both way that $C_3(q,t) = t^3 + qt^2 + qt + q^2t + q^3$



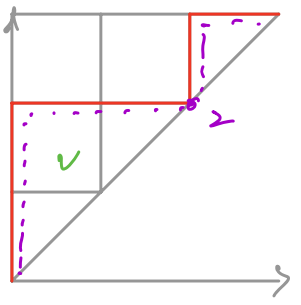
$\text{bounce}(\pi) = 3$

$\text{area}(\pi) = 0$



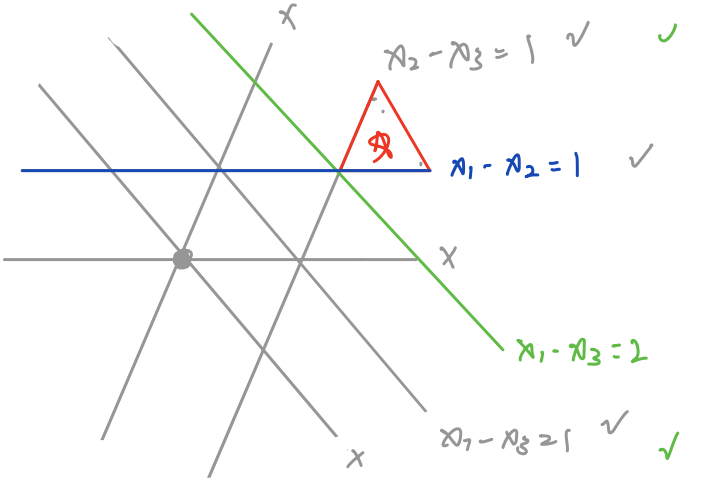
$\text{bounce}(\pi) = 1$

$\text{area}(\pi) = 1$



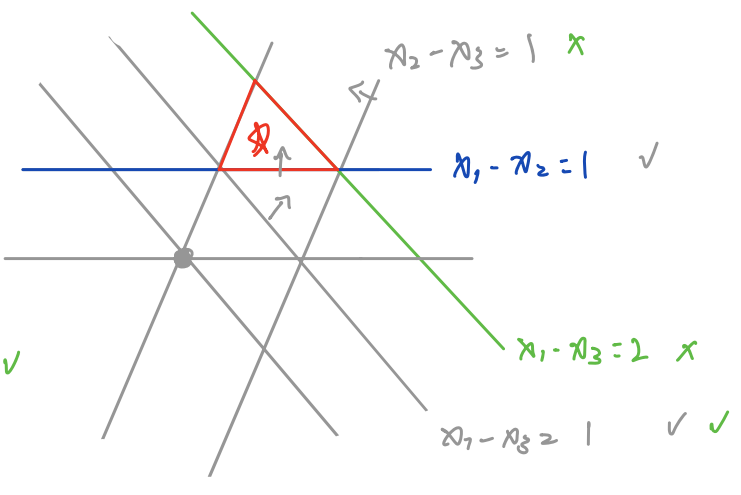
$\text{bounce}(\pi) = 1$

$\text{area}(\pi) = 1$



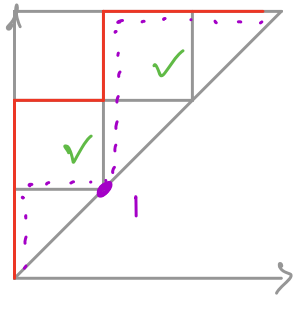
$\binom{3}{2} - \text{sh}(\pi) = 3 - 3 = 0$

$\text{sh}(\pi) = 3$



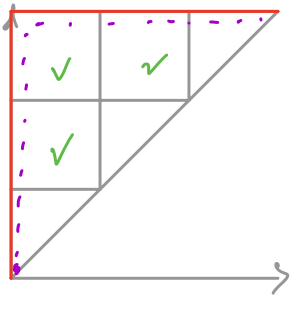
$\binom{3}{2} - \text{sh}(\pi) = 3 - 2 = 1$

$\text{sh}(\pi) = 1$



$\text{bounce}(\pi) = 1$

$\text{area}(\pi) = 2$



$\text{bounce}(\pi) = 0$

$\text{area}(\pi) = 3$

→ correspond to the fundamental alcove.

First class: Hilbert Series on diagonal harmonic

To find combinatorial formula for $\mathcal{H}(DH_n; q, t)$, Haglund, Loehr introduce a statistic "area'", which generalize "area".

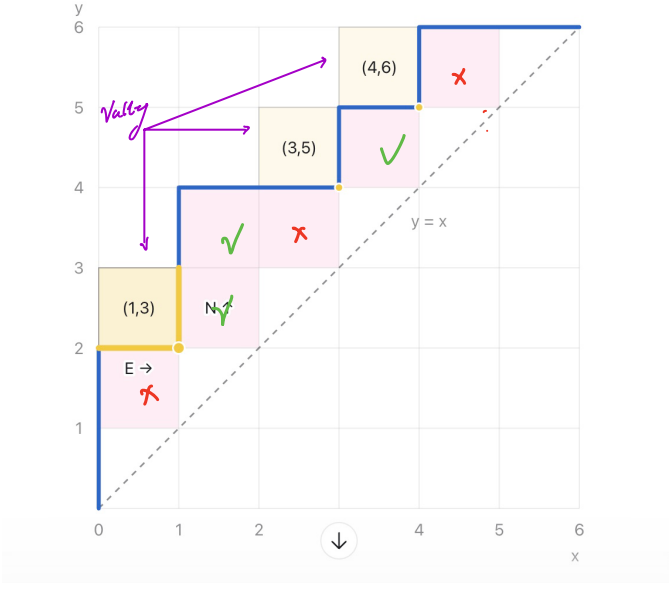
- Let L_n to be the set of (w, π), which satisfies

  (i) $\pi \in L^+_{n,n}$, $w \in \mathfrak{S}_n$.

  (ii) at each valley (i,j), $w(i) \geq w(j)$.

Define

$$\text{area}'(w, \pi) = \#\{(i,j) \in \text{Area}(\pi) : w(i) \geq w(j)\}$$



for example:

$w = (2,1,4,3,6,5)$.

$w(1) \geq w(3)$

$w(3) \geq w(5)$

$w(4) \geq w(6)$.

Then, Haglund, Loehr conjectured that [2002, PPS4C]

$$\mathcal{H}(DH_n; q, t) = \sum_{(w,\pi) \in L_n} q^{\text{area}'(w,\pi)} t^{\text{bounce}(\pi)}.$$

proved as a corollary of the "compositional shuffle conjecture".

[JAMS, Carlsson-Mellit 2015].

First class: statistic on hyperplane arrangement and

$\mathcal{H}(\mathcal{D}\mathcal{A}_n; q, t)$ in terms of hyperplane arrangement.

- In $\mathcal{H}(\mathcal{D}\mathcal{A}_n^{\mathcal{E}}; q, t) = \sum_{\pi \in L_{n,n}^{+}} q^{\text{area}(\pi)} t^{\text{bounce}(\pi)}$ case,

we already have correspondence:

(i) π : Dyck path $\longleftrightarrow$ $\mathcal{I}$: root ideal $\longleftrightarrow$ C: positive Shi chamber

$\longleftrightarrow$ Δ : positive Shi alcove

(ii) $\text{area}(\pi) = \binom{n}{2} - \text{Shi}(\Delta)$, $\text{bounce}(\pi) = \text{ish}(\Delta)$

- Recall that

$$\mathcal{H}(\mathcal{D}\mathcal{A}_n; q, t) = \sum_{(w,\pi) \in \mathcal{L}_n} q^{\text{area}(w,\pi)} t^{\text{bounce}(\pi)}.$$

where $\mathcal{L}_n$ is the set of (w, π), which satisfies

(i) $\pi \in L_{n,n}^{+}$, $w \in \mathfrak{S}_n$.

(ii) at each valley (i,j), $w(i) > w(j)$. (non-inversion).

Hence, sum runs over only positive Shi alcoves is not enough,

we need to run over all Shi alcoves !

- First, we need to find bijection between

all Shi alcoves $\rightleftarrows$ $\mathcal{L}_n$

due to [Stanley 1996, PAMS]

• (floor) For an alcove A, floor of A means the hyperplane that supports a facet of A so that

(i) Do not cross original plane

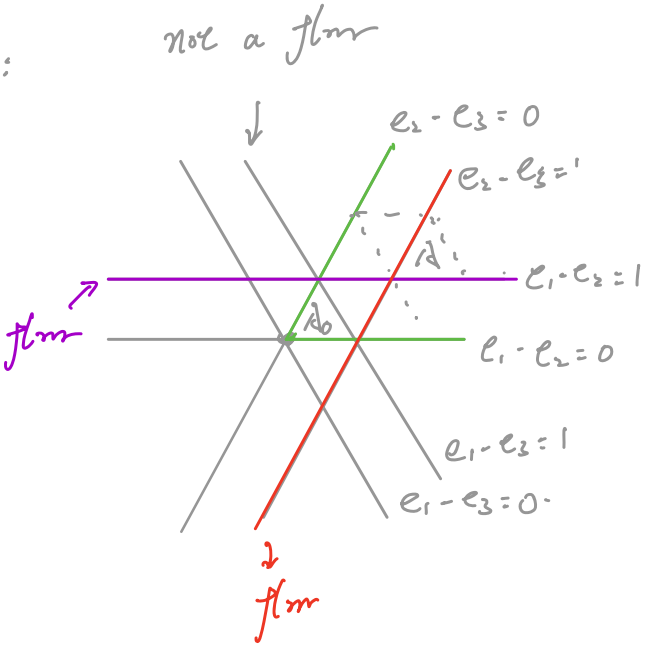
(ii) seperating A_0 and A

• A is a shi alcove if and only if every floor of A belongs to $\text{shi}(n)$. [Shi. 1987]

• A shi hyperplane $x_i - x_j = 1$ is a floor of a positive shi alcove A $\Rightarrow$ $e_i - e_j \in I_A$ is a minimal root

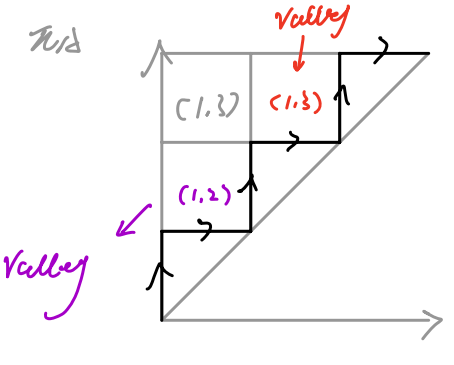
$\Rightarrow$ $(i,j) \in \Pi_A$ is a valley.

Eg:



$I_A = \{ e_1 - e_2, e_2 - e_3, e_1 - e_3 \}$

minimal minimal



The bijection: If A is positive shi alcove, we already have

A: shi alcove $\longrightarrow$ I_A: root ideal

$\longrightarrow$ π_A: Dyck path

If A is non-positive, then, A lie in a unique Coxeter cone wC_0, where $w \in S_n$, C_0 the dominant cone

Then we claim three things

(i): $w^{-1}A \subseteq C_0$ is a positive shi alcove.

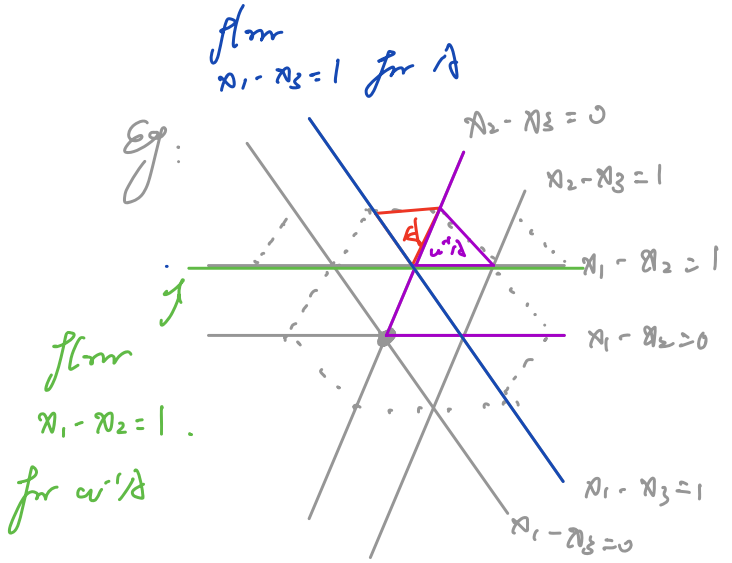
Hence, there is a unique root ideal / Dyck path.

$\pi_{w^{-1}A}$ corresponds to $w^{-1}A$.

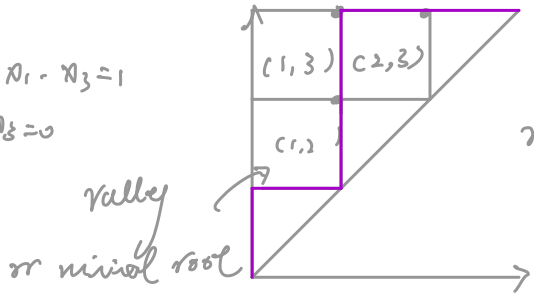
(ii): $(w, \pi_{w^{-1}A}) \in \mathcal{L}_n$.

(iii) For $(w, \pi) \in \mathcal{L}_n$, $\pi \to A \xrightarrow{w} wA$ is a shi alcove.

Eg:



$w = s_2$, $w^{-1}A$ correspond to the ideal $I_{w^{-1}A} = \{e_1 - e_2, e_1 - e_3\}$ and Dyck path $\pi_{w^{-1}A}$:



the valley (1,2), we have

$w(1) = 1$, $w(2) = 3$

$w(1) < w(2)$.

Therefore we have bijection between shi alcoves and $\mathcal{L}_n$.

Two statistics: For a shi-arrangement $\mathcal{A}$ (not necessarily positive),

- $\mathrm{shi}(\mathcal{A}) :=$ the number of shi-hyperplanes that separate $\mathcal{A}$ from $\mathcal{A}_0$.
(same as in the positive case)

- $\mathrm{ish}(\mathcal{A}) :=$ the number of ish-hyperplanes that separate $w\mathcal{A}_0$ from $\mathcal{A}$
where w is the permutation s.t. that $w^{-1}\mathcal{A}$ is positive shi-arrangement.

Then: For a shi-arrangement $\mathcal{A}$, let $(w, \pi) \in \mathcal{L}_n \leftrightarrow \mathcal{A}$. We have

- $\mathrm{area}'(w, \pi) = \mathrm{shi}(\mathcal{A})$
- $\mathrm{bounce}(\pi) = \mathrm{ish}(\mathcal{A})$

Consequently:

$$\mathcal{A}(\mathrm{Dkn}; q, t) = \sum_{\mathcal{A}: \text{shi-arrangement}} q^{\mathrm{shi}(\mathcal{A})} t^{\mathrm{ish}(\mathcal{A})}.$$

End.

q,t-Catalan and hyperplane arrangement

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✓ (i) q,t-Catalan vs hyperplane arrangement

✓ (ii) Hilbert series for diagonal harmonic vs hyperplane arrangement

(iii) Preliminaries for the proof, the proof

(iv) Type B_n q,t-Catalan, Type B_n Shi arrangement, conjectures and open problem.

Main references:

- [Armstrong, 2013, JoC]
- [Haglund, 2004, Lecture notes]
- [Lingxi Lu, 2019, Master thesis]
- [A. Oblomkov, Oct. 2026, arXiv]

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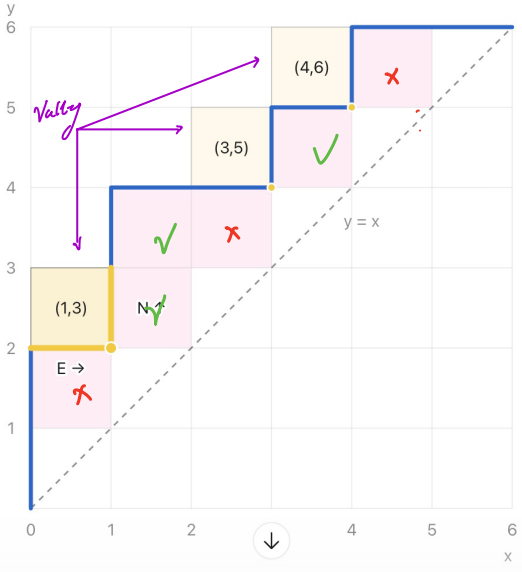
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Second class: $\text{area}(w, \pi) = \binom{n}{2} - \text{shi}(\mathcal{A})$.

Let $\mathcal{A}$ be a shi alcove, $w \in \mathfrak{S}_n$ to be such that $w^{-1}\mathcal{A} \subseteq C_0$, the dominant cone. Let $\pi \in L_n^+$ be the Dyck path corresponds to $w^{-1}\mathcal{A}$. We know that $(w, \pi) \in L_n$, i.e. they satisfy: at each valley (i,j), $w(i) < w(j)$ (non-inversion)

Recall the definition area', it counts the number of squares (i,j) between "π" and "$y = x$" so that $w(i) < w(j)$. i.e. number of non-inversion between "π" and "$y = x$".



Here

$w = (2,1,4,3,6,5)$.

$w(1) < w(3)$

$w(3) < w(5)$

$w(4) < w(6)$.

The proof: First, hyperplane $e_i - e_j = 0$ $(i<j)$ separating $\mathcal{A}$ from $\mathcal{A}_0$ if and only if $\forall x \in \mathcal{A}$, $x_i - x_j < 0$

since $w^{-1}\mathcal{A} \subseteq C_0$, $x_i - x_j < 0$ if and only if $w(i) > w(j)$.

Hence, we have "$e_i - e_j = 0$, $i<j$ separating $\mathcal{A}$ from $\mathcal{A}_0$"

inversion above $y=x$ $\Rightarrow$ "$w(i) > w(j)$, i.e. (i,j) is an inversion of w"

Second, a hyperplane $e_i - e_j = 1$ $(i<j)$ separating A from A_0

if and only if $\forall x \in A$, $x_i - x_j > 1$

if and only if $\forall x \in w^{-1}A$, $x_{w(i)} - x_{w(j)} > 1$ non-inversion above x

if and only if $\begin{cases} w(i) < w(j) \text{ with } e_{w(i)} - e_{w(j)} \in I_{w^{-1}A} \\ w(i) > w(j) : \text{impossible! since } w^{-1}A \subseteq C_0 . \end{cases}$

$x_{w(i)} < x_{w(j)}$ in this case.

Hence: $\operatorname{area}'(w, x)$

$=$ non-inversion between x and $y = x$

$=$ "total cell above $y = x$" $-$ "inversion above $y = x$"

$-$ "non-inversion above x"

$= \binom{n}{2}$ $-$ "hyperplane $e_i - e_j = 0$ separating A from A_0"

$-$ "hyperplane $e_i - e_j = 1$ separating A from A_0"

$= \binom{n}{2} - \operatorname{Shi}(A)$. □

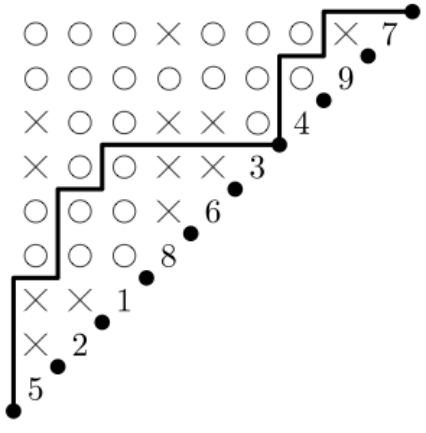


Figure 3.3: A non-positive Shi chamber.

Note: $\times$ stand for inversion of w

$\circ$ stand for non-inversion of w.

Second class: $\text{brnch}(x) = \text{ish}(A)$.

Recall that: $\text{ish}(A)$ counts the number of ish hyperplane that separating $\omega^1 A$ from A_0.

$$\text{Ish}(n) = \{\alpha \in \mathbb{R}^n : \alpha_i - \alpha_n = a : 1 \le i < n,\ a = 1, \dots, n-i\}.$$

Only hyperplanes in $\{\alpha_i - \alpha_n = a : 1 \le i < n,\ a = 1, \dots, n-i\}$ will contribute to $\text{ish}(A)$.

- preliminaries: Address:

  - For $\alpha \in \Phi^+$, $k \in \mathbb{R}$, let $H_{\alpha,k}$ denote the hyperplane $\{x \in V : (x, \alpha) = k\}$. In particular, if $\alpha = e_i - e_j$, then $H_{\alpha,k} = e_i - e_j = k$.

  - For an alcove A and $\alpha \in \Phi^+$, $\exists!\ k \in \mathbb{Z}$, s.t. A lie between $H_{\alpha,k}$ and $H_{\alpha,k+1}$. Hence, for an alcove A, A defines a map

$$k_A : \Phi^+ \longrightarrow \mathbb{Z}.$$

  The function k_A is called the address of A. It gives the precise location of A.

  - Given i, the number of hyperplanes in

$$\{\alpha_i - \alpha_n = a : a \in \mathbb{Z}\}$$

  separating A_0 from A is just $k_A(\alpha)$ $\alpha = e_i - e_n$.

In conclusion, the problem become to prove

$$\text{bounce}(\pi_0) = \sum_{i=1}^{n-1} k_{I_\lambda}(e_i - e_n)$$



· Recall that:

positive chi alcove $A \rightleftarrows$ positive chi chamber $\rightarrow$ ideal I_λ.

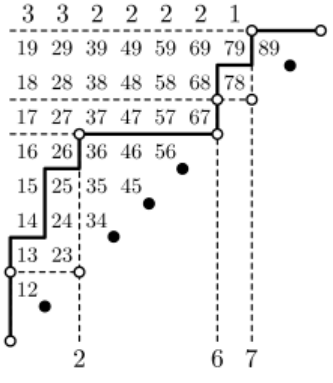
[Sommers, 2005, Canad. Math. Bull, Lemma 3.1, 5.2].

The address of alcove A is given by

$$k_A = k_{I_\lambda} : \mathbb{E}^1 \to \mathbb{Z}$$

$z \mapsto$ the maximum number r so that z can be expressed as a sum of r roots in I

The proof: First notice that, bounce path decompose I_λ into several "blocks", see for example:



Here we have 3 blocks.

Figure 3.4: The bounce path decomposition.

Suppose there are m blocks $B_1, \dots, B_m$ for the longest to the highest. Let $\ell(B_i)$ be the number of cells in the ith row of block B_i. Then $bmax(x)$ pushes the diagonal exactly at the point $(\ell(B_1), \ell(B_2))$. Hence

$$bmax(x) = \sum_{i=1}^{m} \ell(B_i)$$

and it is now sufficient to show that

$$ish(\lambda) = \sum_{i=1}^{n-1} k_{\lambda}(e_i - e_n) = \sum_{i=1}^{m} \ell(B_i) = bmax(x)$$

for the root $\lambda = e_i - e_n$ corresponding to the cell (i, n). Suppose we have $\lambda = r_1 + \cdots + r_r$ where each $r_i \in I_{\lambda}$. Then we can reorder the summands such that $r_j = e_{b_j} - e_{b_{j+1}}$ for some

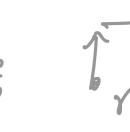
$$1 \le b_1 < b_2 < \cdots < b_{r+1} \le n.$$

Then,

(i) r_{j+1} is above and to the right of r_j

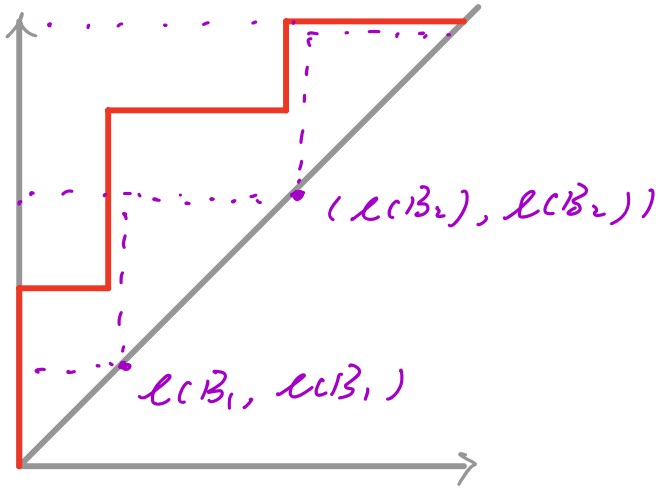
(ii) r_j, r_{j+1} in different blocks

(iii) block that r_j lie in intersect column i.



proof: (i) trivial

(ii) for a block B_k, $(i, j) \in B_k \Rightarrow i \le \ell(B_k) < j$



Now if $r_i \in B_k$, then $b_i \le l(B_k) < b_{i+1} \Rightarrow r_{i+1} \notin B_k$.

(iii): since $d = e_i - e_n = (e_{b_1} - e_{b_2}) + \cdots + (e_{b_r} - e_{b_{r+1}})$

we know $b_1 = i$, and hence each $b_j \ge i$.

Consequently, each block B_k containing r_j satisfies

$$l(B_k) \ge b_j \ge i.$$

This gives "B_k intersect column i"

(i) ~ (iii) together gives:

$$k_A(e_i - e_n) = \max r \le \text{the number of blocks that intersect column } i.$$

We show that they are indeed equal, i.e. let

$$m_i = \text{the number of blocks that intersect column } i.$$

Then, $\exists\ r_1, \ldots, r_{m_i}$, so that $d = e_i - e_n = r_1 + \cdots + r_{m_i}$, and hence $k_A(e_i - e_n) \ge m_i$.

- Construction of $r_1, \ldots, r_{m_i}$: first notice that, if B_k intersect with column i, then B_j, $j \ge k$ also intersect with column i. Hence, B_{n-m_i+1} is the first block that intersect with column i.

Let $r_1 = e_i - e_{l(B_{n-m_i+2})} \in B_{n-m_i+1}$

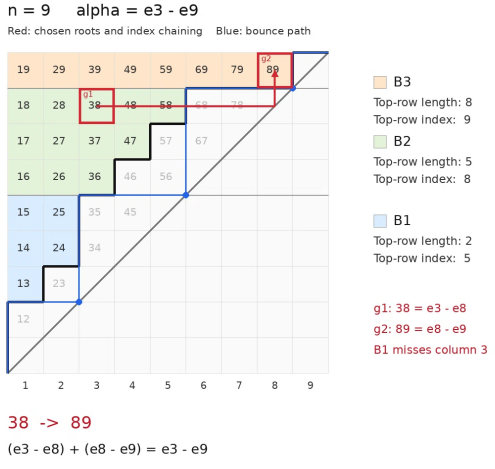
$r_2 = e_{l(B_{n-m_i+2})} - e_{l(B_{n-m_i+3})} \in B_{n-m_i+2}$

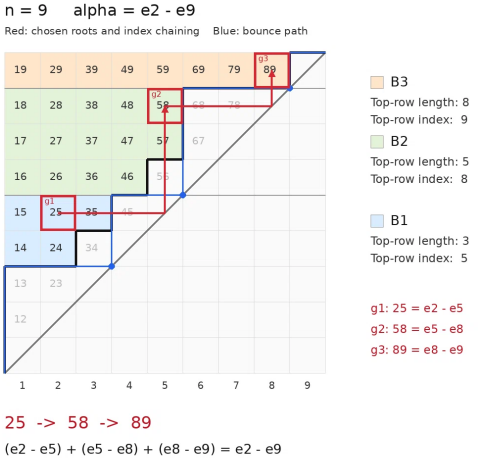
$\vdots$

$(2 \le j < m_{i+1})$ $\quad \gamma_j = e_{\ell(B_u - m_i + j)} - e_{\ell(B_u - m_i + j + 1)} \in B_u - m_i + j$

$\gamma_{m_i} = e_{\ell(B_u)} - e_u \in B_u$

Then $\alpha = e_i - e_u = \gamma_1 + \cdots + \gamma_j$. Done.





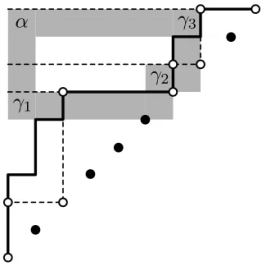


Figure 3.5: The root α as a sum of three roots $\gamma_1 + \gamma_2 + \gamma_3$ above the path.

q,t-Catalan and hyperplane arrangement

This seminar includes four topics

✓ (i) q,t-Catalan vs hyperplane arrangement

✓ (ii) Hilbert series for diagonal harmonic vs hyperplane arrangement

✓ (iii) Preliminaries for the proof, the proof

(iv) Type B_n q,t-Catalan, Type B_n Shi arrangement, conjectures and open problem.

Main references:

- [Armstrong, 2013, JoC]
- [Haglund, 2004, Lecture notes]
- [Lingxi Lu, 2019, Master thesis]
- [A. Oblomkov, Oct. 2026, arXiv]

2026.10.1

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